

Second-Order Model Transfer Function and Impulse Response

We consider one of the canonical forms of an underdamped second-order system transfer function

$$G(s) = K_{dc} \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_d^2} \quad (1)$$

where $\zeta\omega_n$ is the real part of the complex poles and ω_d is the imaginary part of the complex poles. We write it in this form so that the DC-gain of the system is K_{dc} . To illustrate this, recall that $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ so that

$$\lim_{s \rightarrow 0} [G(s)] = K_{dc} \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_d^2} = K_{dc} \frac{(\zeta\omega_n)^2 + \omega_d^2}{(s + \zeta\omega_n)^2 + \omega_d^2} K_{dc}. \quad (2)$$

The impulse response of this model is

$$c(t) = \mathcal{L}^{-1}[G(s)], \quad (3)$$

We use the Laplace transform pair

$$\mathcal{L}[e^{-at} \sin(bt)] = \frac{b}{(s+a)^2 + b^2} \quad (4)$$

applied to (1) to solve for

$$c(t) = \mathcal{L}^{-1} \left[K_{dc} \left(\frac{\omega_n^2}{\omega_d} \right) \frac{\omega_d}{(s + \zeta\omega_n)^2 + \omega_d^2} \right] \quad (5)$$

$$K_{dc} \left(\frac{\omega_n^2}{\omega_d} \right) e^{-\zeta\omega_n t} \sin(\omega_d t) \quad (6)$$

$$K_{dc} \left(\frac{(\zeta\omega_n)^2 + \omega_d^2}{\omega_d} \right) e^{-\zeta\omega_n t} \sin(\omega_d t). \quad (7)$$

For example, consider the system modeled by the transfer function

$$G(s) = K_{dc} \frac{0.5^2 + (2\pi)^2}{(s + 0.5)^2 + (2\pi)^2} \quad (8)$$

which has an impulse response

$$c(t) = K_{dc} \left(\frac{0.5^2 + (2\pi)^2}{2\pi} \right) e^{-0.5t} \sin((2\pi)t) \quad (9)$$